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Highly effective hydrogenation of CO_2 to methanol over $Cu/ZnO/Al_2O_3$ catalyst: A process economy & environmental aspects

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ABSTRACT

The hydrogenation of CO_2 to methanol is one of the promising CO_2 utilization routes in the industry that can contribute to emissions mitigation. In this work, improved operating conditions were reported for the sustainable catalytic hydrogenation of CO_2 to methanol using $Cu/ZnO/Al_2O_3$ catalyst operated at 70 bar and 210 °C. The CO_2 feedstock used for this process is pure CO_2 produced from the cryogenic upgrading process of biogas or hydrocarbon industries and ready-to-use hydrogen purchased at 30 bar and 25 °C. The process was modeled and simulated using the commercial Aspen Plus software to produce methanol with a purity greater than 99% at 1 bar and 25 °C. The simulation results revealed that an adiabatic reactor operated with a CO_2/H_2 ratio of 1:7 produces methanol with a yield ≥99.84% and a CO_2 conversion of 95.66%. Optimizing the heat exchanger network (HEN) achieved energy savings of 63% and reduced total direct and indirect CO_2 emissions by 97.8%. The proposed methanol process with an annual production rate of 2.34 kt/yr is economically sound with a payback period of nine years if the maximum H_2 price remains below $0.97/kg. Hence, producing or purchasing gray H_2 from a steam reforming plant is the most viable economic source for the process.

1. Introduction

Carbon dioxide (CO_2) emissions to the atmosphere exhibit an economic burden and an environmental threat due to their significant contribution to climate change and global warming [1, 2]. In 2021, according to the International Energy Agency (IEA) the global CO_2 emissions were projected to climb by 5% with a total of 1.5 billion tons. This would be the second-highest increase in history and the largest yearly increase in emissions since 2010 [3]. Consequently, different strategies including deploying carbon capture and storage (CCS) and carbon capture and utilization (CCU) were deployed to control the emissions of massive amounts of CO_2 to the atmosphere and to replace depleted fossil fuels in the future. Reusing and recycling CO_2 can contribute to decreasing the effects of global warming and the production of renewable fuels like methanol.

The use of CCS has been a trending topic in industry and literature over the last three decades given the role of this technology in minimizing industrial CO_2 emissions to the atmosphere [4–6]. The CCS is an environmental solution supporting the project's environmental sustainability rather than providing economic value. Consequently, carbon utilization technologies (CCU) have been studied for deployment within oil and gas infrastructures to sustain projects' environmental and economic importance. In CCU, captured and treated CO_2 can be utilized for enhanced oil recovery to increase production or as a feedstock in different industries such as food, chemicals, and fuels. Hence, captured and treated CO_2 can be either liquefied or compressed for direct selling to different sectors or utilized within the same plant by deploying CO_2 monetization technologies to value-added products.

Conversion of CO_2 to methanol has been considered among the most favorable $_{CO2}$ utilization processes in the industry in the last ten to fifteen years due to the maturity and stability of the investigated catalytic systems. [7–10]. Methanol is a liquid chemical that can be used as (i) a solvent; (ii) feedstock for producing chemicals such as acetic acid, methyl *tert*-butyl ether (MTBE), formaldehyde, and dimethyl ether (DME), or (iii) as a cleaner fuel in the transportation sector. In the transportation sector, pure methanol can be used directly as a marine or vehicle fuel, or blended with gasoline for vehicle utilization. Currently, Asia Pacific holds the largest market share of methanol due to the rapid increase in methanol consumption in the automotive, construction, and electronics industries. The global methanol market is projected to reach $26 billion by 2025, with a compound annual growth rate of 6.6% from 2019 to 2025 [11]. Traditionally, methanol has been produced from

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fossil fuels such as natural gas and coal. However, methanol production from captured CO_2 is an emerging route aiming to assure a sustainable production of methanol after the depletion of fossil fuels and to support the efforts to control and mitigate CO_2 emissions.

The feasibility and efficiency of different chemical, electrochemical, and thermochemical reactors for methanol production have been studied in the literature in the past few years. For instance, Kim et al. [12] assessed the energy efficiency and economic feasibility of a solar-based process for the production of methanol from CO_2 and water based on two-step routes. In the proposed process, the first step consists of a thermochemical reactor utilizing solar energy for converting CO_2 to CO using a water gas shift reaction. Synthesis gas consisting of CO and H_2 is then fed to a methanol catalytic reactor for methanol synthesis. The study concluded that the two-step solar-thermochemical pathway is a promising approach for CO_2 utilization to fuels. However, the solar reactor sub-system is capital intensive, and much work must be done to improve the process from an economic perspective. On the other hand, Al-Kalbani et al. [13] modeled and compared the energy assessment of methanol production from CO_2-based chemical vs electrochemical production processes. The authors reported that methanol production based on high-temperature CO_2 electrolysis has double energy efficiency as CO_2 hydrogenation utilizing H_2 produced via water electrolysis. Nonetheless, from an economic perspective, CO_2 hydrogenation to a methanol-based chemical route is found to be the most feasible solution.

Methanol can be produced via catalytic CO_2 hydrogenation using homogenous and heterogeneous catalyst systems wherein the reaction pathway mainly depends on the catalyst. To date, different research studies have investigated or developed the conversion of CO_2 to methanol on a pilot scale using heterogeneous catalysts [14–17]. In 1996, the first commercial low-pressure methanol synthesis process was patented with process conditions below 150 bar and 300 °C using a Cu/Zn-based catalyst [18]. Since then, $Cu/ZnO/Al_2O_3$ catalysts have been widely used in industry and studied in the literature for methanol synthesis due to the superior advantages, where the promotor ZnO provides a high dispersion and stabilization level of Cu active sites and the metal oxide (Al_2O_3) provides support for the catalyst [19,20]; hence, contributing to enhancing methanol production reaction. Different industrially mature catalyst has been studied for CO_2 hydrogenation to methanol from synthesis gas in the temperature and pressure ranges of 210–250 °C and 50–100 bars. In this regard, Van-Dal and Bouallou [21] designed and simulated a CO_2 hydrogenation to methanol plant combined with a CO_2 capture unit and hydrogen production unit using Aspen Plus software. In the CO_2 hydrogenation section of the plant, $Cu/ZnO/Al_2O_3$ catalyst was used in the adiabatic reactor. The steam formed in the methanol synthesis unit was utilized as a CO_2 capture unit. Atsonios et al. [22] investigated a techno-economic analysis of methanol production from CO_2 hydrogenation using a membrane reactor. The study focused mainly on exploring the most economical operation conditions for captured CO_2 utilization to methanol and revealed that hydrogen production costs largely influence the economic feasibility of the process. A thermo-economic approach for methanol production from different renewable sources was proposed by Rivarolo et al. [23]. The authors reported two plant configurations, for which CO_2 is obtained from biogas or purchased from an external plant. Furthermore, the study mainly focused on electricity generation from renewable hydroelectric, wind, or photovoltaic plants and investigated the option of purchasing electricity if renewable resources are not available.

Similarly, Bellotti et al. [8] reported a thermo-economic feasibility study of a power-to-fuel plant for methanol production from CO_2. The studied system consists of a methanol production plant, a hydrogen production plant from water electrolysis, and an amine-based CO_2 capture unit. The authors concluded that selling the by-product O_2 is essential for economic feasibility results. However, neither this study nor the above-mentioned studies have considered the proposed plant's design and simulation. Other studies in the literature either focused on the reaction pathway and kinetics of CO_2 hydrogenation to methanol over $Cu/ZnO/Al_2O_3$ [24–28], or elaborated on catalyst preparation, catalyst formulation, and reaction mechanisms for CO_2 hydrogenation to methanol over different catalysts [29–36]. To the best of the authors' knowledge, limited studies in the literature provided a comprehensive analysis of the CO_2 hydrogenation process to methanol, wherein the majority of these studies focused on thermo-economic aspects of plant configuration or the feasibility of employing different technologies to support the feasibility of methanol production. Moreover, operating conditions such as temperature, pressure and H_2/O_2 feed ratio, and reactor types have not been intensively examined in previous studies. Consequently, The purpose of this work is to assess the technoeconomic-environmental feasibility of CO_2 hydrogenation to the methanol process using the commercial catalyst, $Cu/ZnO/Al_2O_3$, with updated operating conditions for improved CO_2 conversion. The methanol synthesis process is modeled and simulated using the commercial software Aspen Plus V11. Both isothermal and adiabatic reactors are studied for methanol synthesis under fixed feed conditions and catalyst specifications. Additionally, a sensitivity analysis is considered to investigate the influence of temperature, pressure, and variable hydrogen feed on the methanol yield. The optimized process is finally evaluated under environmental and economic aspects. In comparison with other studies that consider captured CO_2 utilization for methanol production, this study emphasizes on the practicality and profitability of deploying CO_2 to methanol monetization infrastructure within the biomass value chain for direct utilization of liquid pure CO_2 by-product produced from a cryogenic biogas upgrade process. The proposed process could also be employed for the CO_2 utilization from petrochemical processes where the raw materials (CO_2 and H_2) are available.

2. Process description

The proposed process utilizes 76.46 kmol/hr of CO_2 by-product produced from a cryogenic biogas upgrading process at 12.3 °C and 47.63 bar within the same plant, and 535.22 kmol/hr of hydrogen supplied at 25 °C and 30 bar. Depending on the plant capacity and purchase price of hydrogen, hydrogen can be supplied from a renewable source such as water electrolysis, or a fossil-fuel resource such as natural gas or coal. The deployment of a CO_2 monetization process to value-added methanol within the biogas value chain achieves two main targets: (1) minimizing CO_2 emissions, and (2) enhancing the economic performance of the biogas value chain. The proposed configuration of the full-scale biogas upgrading process is illustrated in Fig. 1. Although the proposed CO_2 hydrogenation to methanol is connected to the biogas process, it could also be employed for the CO_2 utilization from petrochemical processes where the required raw materials (CO_2 and H_2) are available.

Fig. 2 presents the process used for CO_2 hydrogenation to methanol. The process consists of a feed preparation section to meet the required conversion conditions; a reactor section where catalytic CO_2 conversion takes place and a purification section to produce methanol with purity $\geq$ 98%. The methodology for modeling the methanol synthesis process was mainly inspired by Van-Dal and Bouallou [21]. For designing and simulating the CO_2 hydrogenation process in Aspen Plus, Redlich-Kwong (RKS) equation of state can be optimally used to simulate the process kinetics as reported previously in the literature [37,38]. In contrast, other studies reported using different equations of states, such as Non-Random Two Liquid (NRTL) [7], or employed more than one equation of state, such as RKSMHV2 and NRTL-RK based on the stream pressure [21].

The following is a thorough description of each of the sections in this process.

2.1. Feed preparation:

In the first section of the conversion process, the feeds (CO_2 and H_2) are compressed to 78 bar, mixed, and heated to 210 °C to meet the

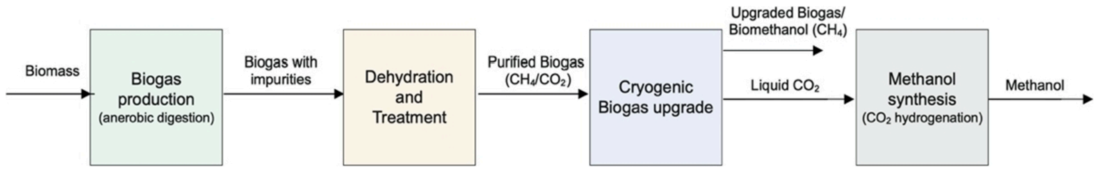

Fig. 1. Block flow diagram of the full-scale biogas upgrading process.

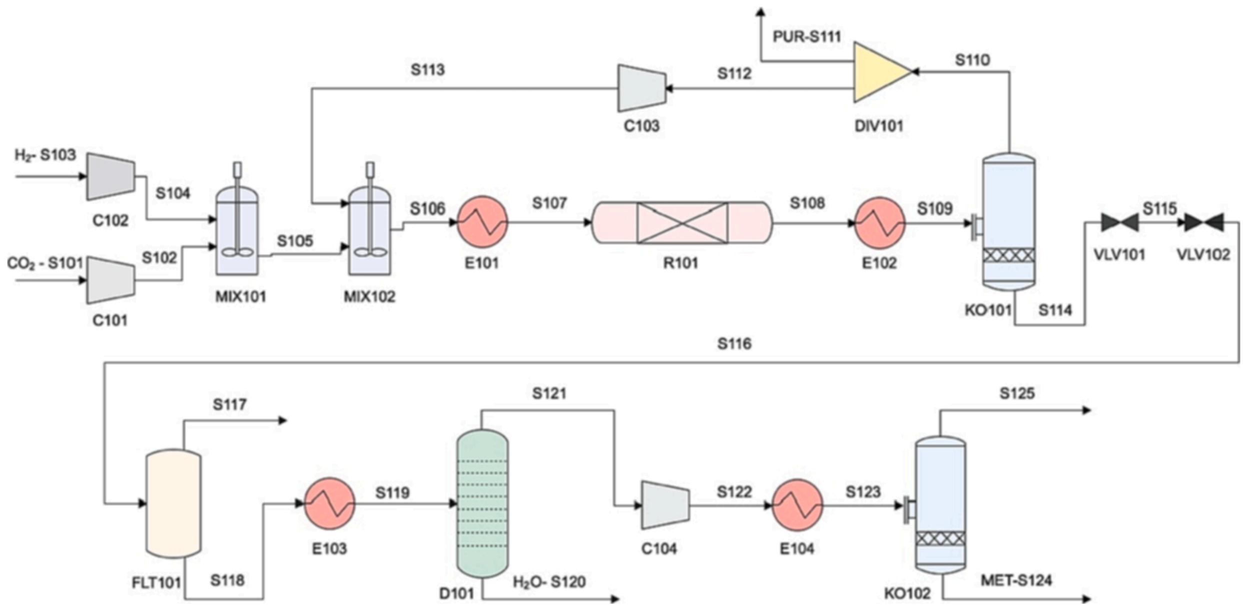

Fig. 2. flow diagram of the proposed CO_2 to methanol conversion process.

reactor inlet specifications. An advantage of utilizing CO_2 produced from the cryogenic biogas upgrade process is that the produced CO_2 is at high pressure and does not require high power for compressing compared to other proposed processes in the literature where multi-stage compressors were used [21,37,39,40]. Additionally, the CO_2 is supplied with high purity and does not require any treatment before feeding into the conversion process. If CO_2 is supplied from other industries the treatment and compression work should be considered.

2.2. Reaction section:

In this section, CO_2 is mixed with /H_2 then compress to 75.7 bar and heated to 210 °C, and fed to the fixed bed plug flow reactor to produce methanol. Adiabatic and isothermal operating conditions were tested in this study. The reactor system contains 44,500 kg of $Cu/ZnO/Al_2O_3$ catalyst. The properties of the $Cu/ZnO/Al_2O_3$ are summarized in Table 1.

Two parallel exothermic reactions Eq. (1) and (2) take place inside the reactor to produce methanol alone with an endothermic reverse-water gas shift (RWGS) reaction Eq. (3):

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH\,(l) + H_2O_g \Delta H = -87\,KJ/mol(25^\circ C) \quad (1)$$

Table 1
Characteristics of the catalyst [21].

$Cu/ZnO/Al_2O_3$	
Density	1775 Kg_{cat}/m^3_{cat}
Fixed bed porosity	0.5
Mass	34.8 g
Pellet diameter	0.0005 *m*

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH\,(l) \Delta H = -128\,KJ/mol(25^\circ C) \quad (2)$$

$$CO_2(g) + H_2(g) \rightleftharpoons CO_g + H_2O_g \Delta H = 41\,KJ/mol(25^\circ C) \quad (3)$$

In the presence of the catalyst, the used kinetic model is based on Langmuir-Hinshelwood- Hougen- Waston (LHHW) mechanism and assumes CO_2 as the primary source for methanol production in the presence of the RWGS reaction [41]. The kinetic model parameters were further modified by Mignard and Pritchard [42] to application ranges up to 75 bar as shown in Eq. (4) and (5), where the pressure is in bar and temperature in K. The kinetic constants, Eq. (6), follow the Arrhenius law, while the thermodynamic equilibrium constants, Eq. (7) and (8), are given by Graaf et al. [43]:

$$r_{CH_3OH} = \frac{k_1 P_{CO_2} P_{H_2}\left(1 - \frac{1}{K_{eq2}} \frac{P_{H_2O} P_{CH_3OH}}{P_{H_2}^3 P_{CO_2}}\right)}{\left(1 + k_2 \frac{P_{H2O}}{P_{H_2}} + k_3 P_{H_2}^{0.5} + k_4 P_{H_2O}\right)^3} \left[\frac{mol}{kg_{cat} s}\right] \quad (4)$$

$$r_{RWGS} = \frac{k_5 P_{CO_2}\left(1 - \frac{1}{K_{eq1}} \frac{P_{H_2O} P_{CO}}{P_{CO_2} P_{H_2}}\right)}{\left(1 + k_2 \frac{P_{H2O}}{P_{H_2}} + k_3 P_{H_2}^{0.5} + k_4 P_{H_2O}\right)} \left[\frac{mol}{kg_{cat} s}\right] \quad (5)$$

$$k_i = A_i \exp\left(\frac{B_i}{RT}\right) \quad (6)$$

$$\log_{10}\frac{1}{K_{eq1}} = \frac{2073}{T} + 2.029 \quad (7)$$

$$\log_{10} K_{eq2} = \frac{3066}{T} - 10.592 \quad (8)$$

The equations(1 to 8) were rearranged in alignment with the type of accepted kinetic equations in Aspen Plus software and represented in equations (9 to 11). Table 2 summarizes the model parameters used in the Aspen Plus software [21].

$$r_{CH_3OH} = \frac{k_5 P_{CO_2} - k_6 P_{H_2O} P_{CH_3OH} P_{H_2}^{-2}}{\left(1 + k_2 P_{H_2O} P_{H_2}^{-1} + k_3 P_{H_2}^{0.5} + k_4 P_{H_2O}\right)^3} \quad \left[\frac{mol}{kg_{cat} s}\right] \tag{9}$$

$$r_{RWGS} = \frac{k_5 P_{CO_2} - k_7 P_{H_2O} P_{CO} P_{H_2}^{-1}}{1 + k_2 P_{H_2O} P_{H_2}^{-1} + k_3 P_{H_2}^{0.5} + k_4 P_{H_2O}} \quad \left[\frac{mol}{kg_{cat} s}\right] \tag{10}$$

$$\ln k_i = A_i + \frac{B_i}{T} \tag{11}$$

Additionally, a multi-tube reactor (# of tubes = 1000 tubes, length = 5 m, and diameter = 1 m) was considered. A pressure drop of 0.6 bar is allowed through the reactor, with an outlet stream leaving the reactor at 75 bar. Exact specifications are applied for the isothermal reactor, with a constant reactor operating temperature of 210 °C.

2.3. Purification section:

Gases leaving the reaction were collected in the knock-out drum (KO101) to separate the products from unreacted reactants. Unreacted gases are recycled back to the reactor to enhance conversion and part of it is purged to the atmosphere at a split fraction of 0.1 to avoid by-product accumulation. The produced liquid methanol leaves the knock-out drum at 73.4 bar. Two parallel valves are used to reduce its pressure down to 1.2 bar before entering the flash drum (FLT101) for further purifications and removal of unreacted gases. The outlet liquid methanol from the flash drum at 1.2 bar and 14.95 °C is heated up to 80 °C before sending it to the distillation column (D101) for methanol/water separation. A distillation column with 15 stages and a reflux ratio of 2.12 is utilized to produce a high-grade methanol product at 64.92 °C and 1 bar, and water by-product at 101.91 °C and 1 bar. No pressure drop is assumed across the distillation column. The produced methanol is then compressed to 80.29 bar, heated to 80.29 °C, and fed to a final knock-drum (KO102) to increase the purity of the methanol to 99.41 mol %.

In this process, all compressors are isentropic and operate at 72% efficiency. Moreover, a stream pressure drop between 0.1 and 2.3 bar is allowed in the heat exchangers. Table 3 presents the main specifications of the designed methanol synthesis process.

3. Results and Discussion:

3.1. CO_2 hydrogenation to methanol

Fig. 3 illustrates the simulated CO_2 to methanol process using Aspen Plus. The proposed simulated process utilizes high-pressure pure CO_2 obtained from a cryogenic biogas separation unit within the same plant.

Table 2
Parameters of the rearranged kinetic model [21].

Kinetic model parameters for Aspen Plus		
k_1	**A_1**	−29.87
—	**B_1**	4811.2
k_2	**A_2**	8.147
	B_2	0
k_3	**A_3**	−6.452
	B_3	2068.4
k_4	**A_4**	−34.95
	B_4	14,928.9
k_5	**A_5**	4.804
	B_5	−11,797.5
k_6	**A_6**	17.55
	B_6	−2249.8
k_7	**A_7**	0.1310
	B_7	−7023.5

Table 3
A summary of key process specifications for CO_2 hydrogenation to methanol.

CO_2 hydrogenation to methanol specified process conditions for Aspen Plus	
CO_2 feed temperature and pressure	12.3 °C and 47.63 bar
H_2 feed temperature and pressure	25 °C and 20 bar
CO_2/H_2 ratio	1:7
Reactor inlet temperature and pressure	210 °C and 75.7 bar
Compressors isentropic efficiency	72%
Number of stages in the distillation column	15
Reflux ratio in the distillation column	2.12
Pressure drop in heat Exchangers	Variable pressure drop between 0.1 and 2.3 bar
Stream pressures drop across the reactor	0.6 bar

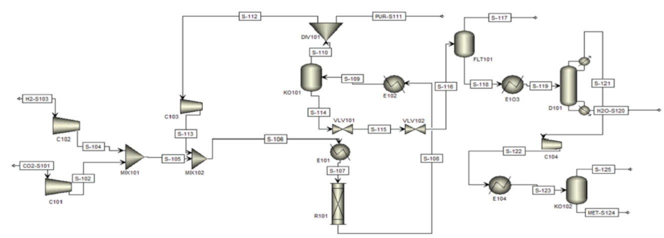

Fig. 3. CO_2 hydrogenation process to methanol by Aspen Plus.

As justified by Rivarolo et al. [23], we are utilizing CO_2 produced from biogas, which offers more excellent economic performance and a more straightforward plant layout. The use of pure and high-pressure CO_2 for methanol production reduces the project's overall costs due to: (1) the employment of a single CO_2 compressor, and (2) limited need for CO_2 collection and purification devices. Hydrogen was assumed to be purchased from a local market to supply the process.

The hydrogenation of CO_2 to methanol takes place in the catalytic reactor (R101). The reactor uses a commercial $Cu/ZnO/Al_2O_3$ as a catalyst. Consequently, designing and modeling the feed preparation section and the separation section mainly depend on the reactor's feed specifications and the composition of the outlet stream. The results revealed that feeding the reactor with CO_2/H_2 mixture at 210 °C and 75 bar achieved CO_2 conversion of 99% and a methanol yield of 98%. The productivity of methanol and the conversion of CO_2 are enhanced by recycling part of the unreacted gas mixture. This is in good agreement with the results reported by Leonzio et al., [44]. Although the recycled stream of gases contains CO, at lower feed gas temperatures, methanol synthesis from CO_2/H_2 is faster than CO/H_2 [45]. Based on Skrzypek et al. [24] who studied methanol synthesis kinetics over $Cu/ZnO/Al_2O_3$ catalyst in a high-pressure fixed bed plug flow reactor, the authors concluded that the surface reaction between CO_2 and H_2 is the rate-controlling step. The authors further reported that the selectivity is higher for a feed that consists of only CO_2 and H_2 without any CO. This reveals that CO_2 is the primary source for methanol synthesis in the process [24,46]. It was observed that increasing the feed pressure up to 75.7 bar significantly improved methanol production and achieved overall CO_2 conversion $\geq$ 99%. This was aimed at favorable operating conditions of the forward reaction following Chatelier's principle. Kiss et al. [7] simulated the process at 50 bar, which resulted in 100% process conversion using Cu/Zn/Al/Zr catalyst, while Atsonios et al. [22] simulated the process at 65 bar, using a membrane reactor and $Cu/ZnO/Al_2O_3$ catalyst, which resulted in 30.5% CO_2 conversion. Table 4 summarizes the main specifications of the inlet and outlet streams of the proposed methanol simulation process.

Additionally, Table 5 compares the CO_2 conversions and methanol yields and/or selectivity of the processes reported by different authors in the literature using adiabatic/isothermal fixed bed flow reactors packed with Cu/Zn or Cu/ZnO_2 based catalysts. As indicated in Table 5, the

Table 4
Summary of the process streams for the simulated CO_2 to methanol process.

Stream	TR (°C)	PR (bar)	Flowrate (kmol/hr)	Characteristics
CO_2-S101	12.3	47.63	76.46	Inlet
H_2-S103	25	30	535.22	Inlet
PUR-S111	35	73.4	583.98	Outlet
S-117	14.95	1.2	0.0097	Outlet
MET-S122	40	1	75.52	Main Product
H_2O-S120	101.92	1	75.52	By-Product

current proposed process's reported CO_2 conversion and methanol yield are higher than other studies and relatively close to the results reported by Kiss et al. [7] carried out in an isothermal plug flow reactor operated at 50 bar and 250 °C. Nevertheless, other authors used the fibrous Cu/Zn/Al/Zr catalyst rather than the industrially mature Cu/ZnO/Al_2O_3 catalyst employed in this study. The obtained results highlight the importance of additional research to demonstrate the impact of reactor type on process conversion and to determine the optimal reactor configuration and operating conditions.

3.2. Reactor Analysis:

Different studies studied the conversion of CO_2 to methanol in adiabatic and isothermal reaction system [7,21,47]. Consequently, the productivity and the conversion of CO_2 in both reactor types were explored and simulated in this study. Tests were performed at 210 °C and 75.8 bar using same previous flow rate and catalyst loading. The obtained results are illustrated in Table 6. The adiabatic reactor produced slightly less CO_2 but had a higher methanol yield and selectivity. The required heat duty and residence time differed significantly between the two reactors. It was observed that the residence of the adiabatic reactor is 50% less than the isothermal reactor. Confirming that the adiabatic reactor system has more favorable operating conditions. Therefore, the adiabatic reactor was subjected to further analysis to understand the effect of operating conditions (Temperature and pressure) and the molar flow rate of H_2 on methanol production and CO_2 conversion. It was proved that methanol production is independent of the reactor temperature at various reactor temperatures due to the low activation energy, which was almost zero. Hence, the changing temperature in the Arrhenius equation does not influence the reaction kinetics.

The influence of changing adiabatic reactor pressure and H_2 molar flow rate on the methanol production is presented in Fig. 4a and b. Results indicated that increasing the adiabatic reactor pressure is directly proportional to methanol conversion. Maximum pressure of 75 bar can be set as the operating pressure due to kinetic limitations. Moreover, increasing the H_2 feed flow rate resulted in higher methanol yield, where a CO_2/H_2 ratio of 1:7 was optimally selected for the study. As can be seen in Fig. 4 (b), increasing the H_2 flow rate beyond 535.22 kmol/hr does not significantly influence the yield, wherein expanding the molar flow rate by 34.6% would only enhance methanol yield by 4.5%.

On the other hand, changing the temperature and/or the pressure of the feed disturbs the flash specifications. Hence optimal feed specifications of 210 °C and 75.8 bar were selected to achieve a CO_2 conversion of 95.66% in an adiabatic reactor, which is relatively lower than the reported CO_2 conversion in the literature. Further optimization of the overall process has been studied to investigate the optimal configuration.

4. Heat exchange network

The presented process involves different endothermic and exothermic units where heat integration is crucial for improved process efficiency. The pinch analysis method proposed by Linnhoff and Hindmarsh [48] was considered for designing an optimal heat exchanger network (HEN) to (1) improve the overall process energy efficiency; (2) minimize the operational costs and utility consumption, and (3) minimize indirect CO_2 emissions due to reduction of fuel consumption for steam generation. The commercial software Aspen Energy Analyzer V11 was used to conduct the pinch analysis where a minimum temperature difference (ΔT_{min}) of 5 °C was selected. The main trade-off when considering a low ΔT_{min} in pinch analysis is between energy/external utility reduction and increased capital costs due to extra heat exchangers. A low ΔT_{min} decreases utility costs but increases the capital costs for installing additional units.

Consequently, the payback time on capital investment was also considered for evaluating the optimal heat integration scenario [49]. The optimal results from the Aspen Energy analyzer were then transferred to the primary Aspen Plus simulation for an updated process flow diagram after heat integration, illustrated in Fig. 5. The implementation of heat integration resulted in 63.19% energy savings after introducing three additional units, RE101, RE102, and RE103. A comparison of the required external utility requirement under optimized HEN vs total utility requirement in the absence of optimized HEN is illustrated in Fig. 6. Results after heat integration imply that external hot utilities were reduced from 3.13 to 0.014 GW achieving more than 99.55% of saving. In addition, the external cooling utilities requirements were reduced from 3.62 to 2.11 GW achieving around 41.7% saving. Consequently, deploying an optimized methanol production will significantly reduce the costs associated with utilities purchasing and/or generation.

5. Economic analysis

The utilization of high-pressure pure CO_2 in the proposed process contributes to reducing both capital costs (Capex) and operating costs (Opex) compared to other models reported in the literature. Under optimized conditions, it was noticed that the proposed process required pure H_2 at a pressure of 30 bar and a total specific power of 214 kWh/

Table 6
Summary of the CO_2 conversion to methanol under isothermal and adiabatic conditions.

	Isothermal	Adiabatic
CO_2 Conv. (%)	96.26	95.66
MeOH Yield (%)	98.84	99.84
MeOH Selectivity (%)	98.85	98.91
Heat duty (KJ/hr)	−46.46 E + 5	20.94
Residence time (hr)	1.102	0.568

Table 5
Comparison of the proposed CO_2 to methanol process with previous studies.

Reference	CO_2/H_2 ratio	Reactor type	TR (°C)	PR (bar)	CO_2 Conv. (%)	Selectivity (%)	Yield (%)
Current study	1:7	Adiabatic	210	75.8	99.06	99.13	98.28
[21]	1:3	Adiabatic	210	78	–	33	–
[37]	1:3	Adiabatic	210	76	94	21	–
[7]	2.84:3	Isothermal	250	50	100	–	99.83
[22]	1:3	–	250	65	30.5	–	–

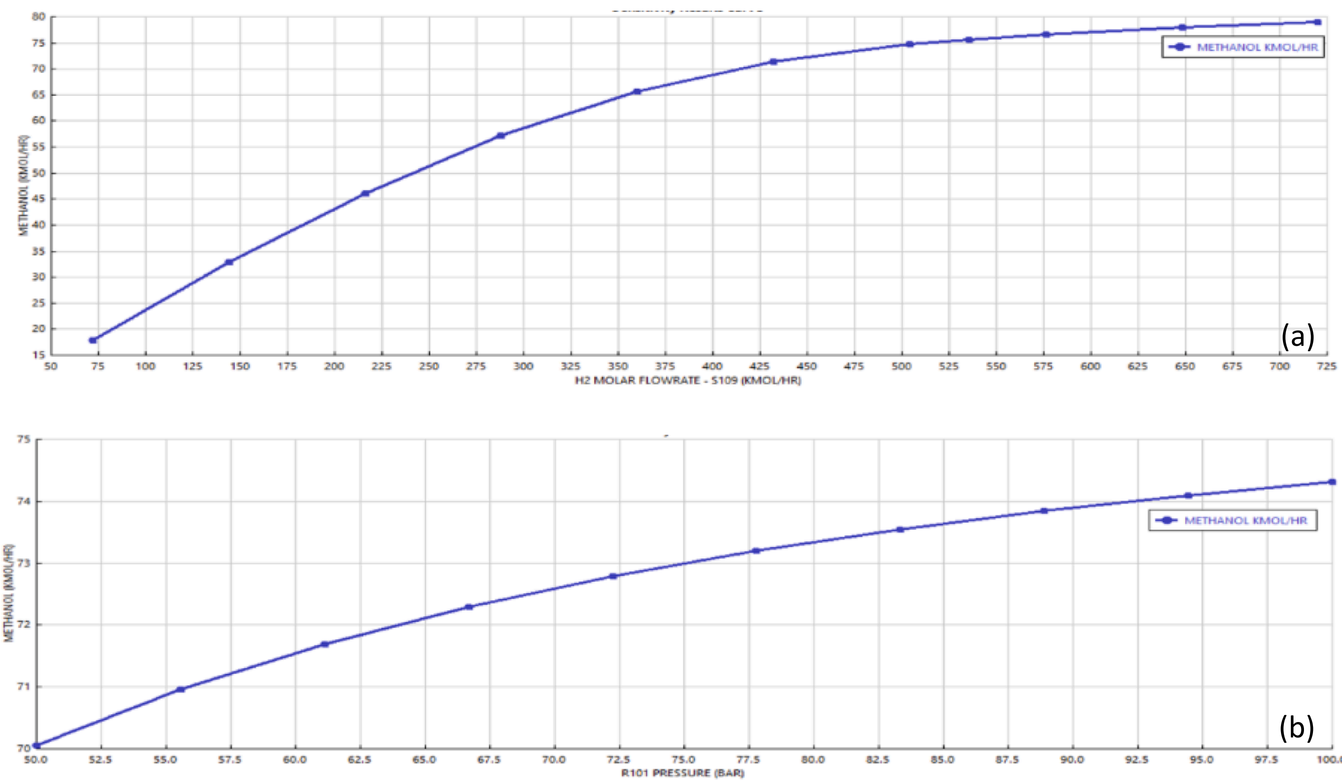

Fig. 4. Methanol yield (kmol/hr) at: (a) varied H_2 molar flow rate in (kmol/hr) and (b) reactor pressure in (bar).

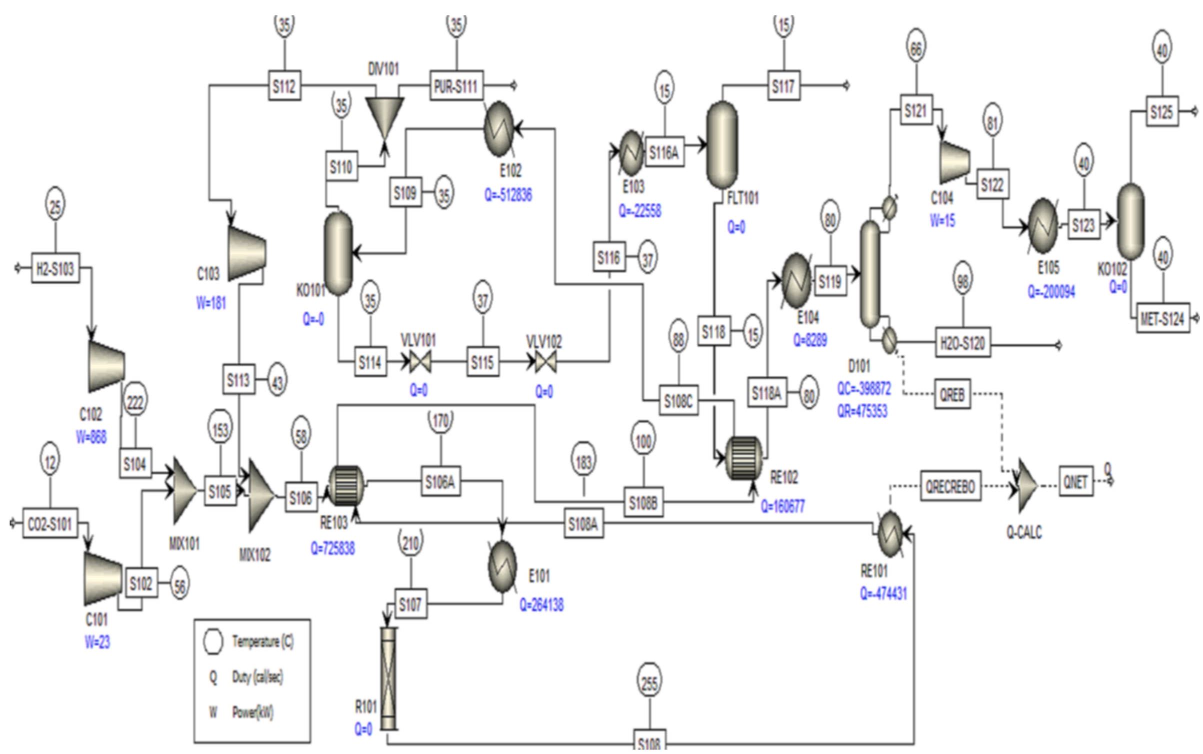

Fig. 5. Optimized process flowsheet for CO_2 to methanol process.

t_{MeOH} to produce methanol with a purity of 99.41 mol%. This is economically feasible if compared with an operating pressure of 65 bar and specific power consumption of 113 kWh/t_{MeOH} to produce methanol with a purity of 99.3 mol% methanol as reported by Atsonios et al. [50]. The reported value did not consider the power requirement for CO_2 feed preparation as it was considered as part of the CO_2 capture and treatment unit in the plant.

Optimizing the CO_2 conversion reduced total utility requirements by 63%, indicating that this process has the potential to generate additional revenue. The plant's economic viability depends on different factors,

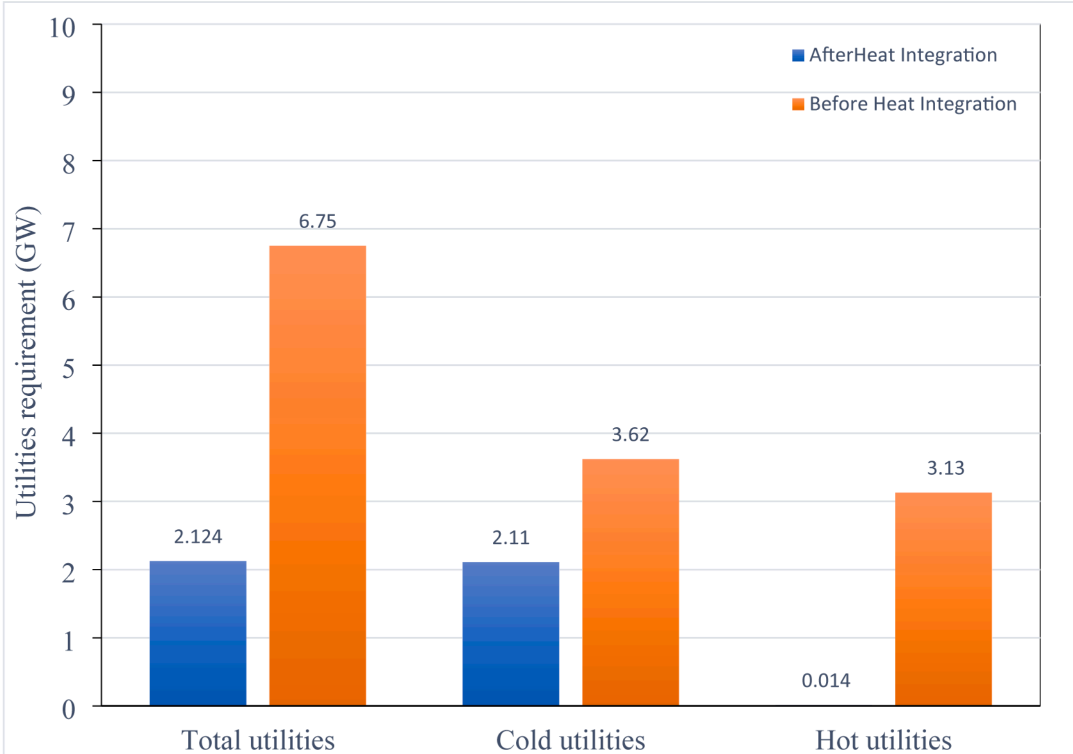

Fig. 6. Required external utility before and after the optimized HEN.

including Capex, utility costs, electricity costs, the project's lifetime, CO_2 taxes in some countries, and methanol price in international markets. The net present value (NPV), equation (12), was used to assist the economy of the methanol production process and determine its profitability:

$$NPV = \sum_{t=1}^{n} \frac{CF_t}{(1+i)^t} \tag{12}$$

where CF represents annual cash flow at any time (t); n is the service life of the project and i is the rate of return on the investment. The profitability of the methanol production process was based on a plant service life of 20 years and a rate of return on investment of 8%. The Capex of the process was determined using the step counting method following the procedure established by Timm's correlation for similar gas processes [51]:

$$Capex = 13000NQ^{0.615} \tag{13}$$

where Capex is in US Dollars for 1998, N is the number of significant processing units, and Q is the annual plant capacity in metric tons (mt). When counting the significant processing units, only reactors, distillation columns, and compressors are considered to have substantial costs [52]. Moreover, since Timm's correlation results in Capex were conducted in 1998, cost indices were used to adjust the Capex value to the year 2021 [53].

Both fixed and variable Opex must be addressed when estimating the Opex. In this analysis, fixed operating and maintenance costs were taken as 1.04% of Capex assumed previously by Bellotti et al., [8]. On the other hand, variable Opex relays on the production capacity, utility requirement, and fuel and electricity costs for running equipment and generating utilities. All compressors are electrically driven and purchased from an external local supplier in Qatar at a \$0.036/kWh [54]. The steam generation total cost (\$/lb_{steam}) was calculated using Eq. (14) [52]:

$$\text{Steam cost} = \text{Fuel Price} \times \frac{\text{Heating rate}}{\text{Boiler efficiency}} \tag{14}$$

where fuel price was taken as a fixed average monthly Henry Hub natural gas price in 2021 of \$3.62/MMBTU; the heating rate is the amount of energy needed to heat feed water to saturated low or high-pressure steam in (Btu/ lb_{steam}) [55], and boilers efficiency of a fixed value of 85.7% was considered [52,56].

For cold utilities, sea water and chilled water were considered for cooling process streams down to 35 and 15 °C, respectively. However, the costs of raw water, makeup water, condensate return, water treatment, and power for pumping cooling water were not analyzed. Further detailed calculations can be considered when assessing the plant on a tactical level of project planning. For example, seawater can be used to cool down process streams down to 35 °C, and chilled water can be used to cool down the process stream (S116) entering FT101 to 15 °C.

The revenues of the proposed CO_2 conversion plant are based on selling the produced liquid methanol at an average price of \$692/mt [57]. The process profitability was supported by the availability of CO_2 from nearby cryogenic biogas or petrochemical processes and the presence of gray H_2. This latter is produced from steam methane reforming in the Middle East and supplied for \$0.9/kg [58]. The NPV and payback period of the methanol plant with a production capacity of 23.4 kt/yr was determined to be \$6.5 million and nine years, respectively based on 20 years of service life.

It is difficult to identify the most competitive methanol production scheme. The economic performance depends on the electricity and/or fuel prices, hydrogen costs, and methanol selling price in international markets. Under fixed Opex and hydrogen supply costs, the profitability of the investment in CO_2 hydrogenation to methanol process was investigated based on different methanol production capacities: 23.4 kt/yr, 33.6 kt/yr, and 44.8kt/yr for the project's lifetime of 20 and 25 years. As shown in Fig. 7, increasing the production capacity up to 3.36 kt/yr and 4.48 kt/yr for a project's lifetime of 20 years results in enhancing the NPV of the project by 147% and 411%, respectively. The NPV is

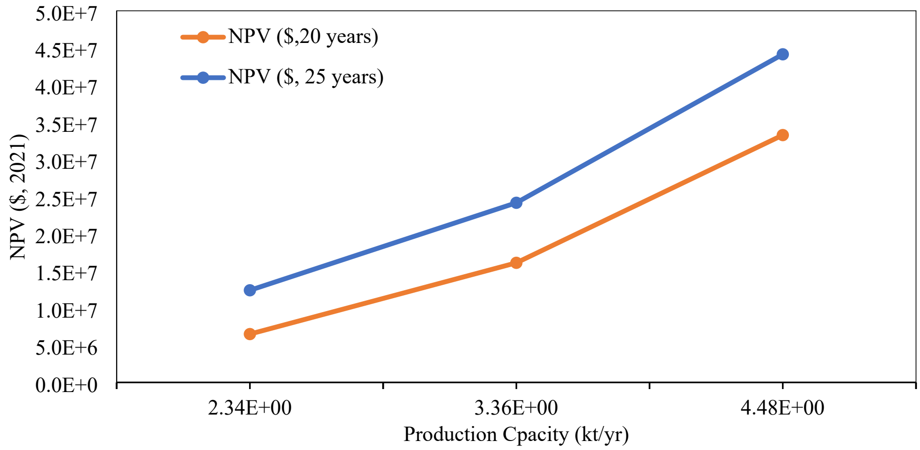

Fig. 7. NPV of CO_2 hydrogenation to methanol process at different production capacities and project lifetime of 20 and 25 years.

further enhanced for all production scenarios when extending the project's lifetime by five additional years. This reflects the economic attractiveness of deploying a bio-methanol process due to the low Opex and the availability of affordable gray H_2 supplied from local steam gas reforming processes. It is worth observing that due to the high requirement of H_2 to satisfy the CO_2/H_2 ratio of 1:7 in the process, a maximum H_2 supply price of $0.97/kg is required to break even the NPV for a 20-year project with annual methanol production of 2.34 kt/yr. Consequently, supplying renewable hydrogen from PEM electrolysis at a price between $4.2/kg and $5.2/kg will be economically infeasible for the proposed CO_2 hydrogenation to methanol process [58].

6. Environmental analysis

As shown in the proposed bio-methanol production process, CO_2 emissions can be released into the atmosphere directly from the main process equipment or indirectly due to burning fuel for generating thermal energy and/or electricity (in case generated locally). The process releases three streams to the atmosphere containing CO_2 for direct emissions: PUR-S111, S-117, and S-125. After heat integration, only burning fuel for generating low-pressure steam contributes to indirect CO_2 emissions to the atmosphere. Consequently, the indirect emissions are mainly influenced by the fuel needed to generate steam utilized in the process. According to the US Energy Information Administration [59], burning 1 MMBtu of natural gas emits around 117 lb of CO_2. A comparison between direct and indirect CO_2 emissions before and after heat integration is presented in Fig. 8. The figure shows that the indirect

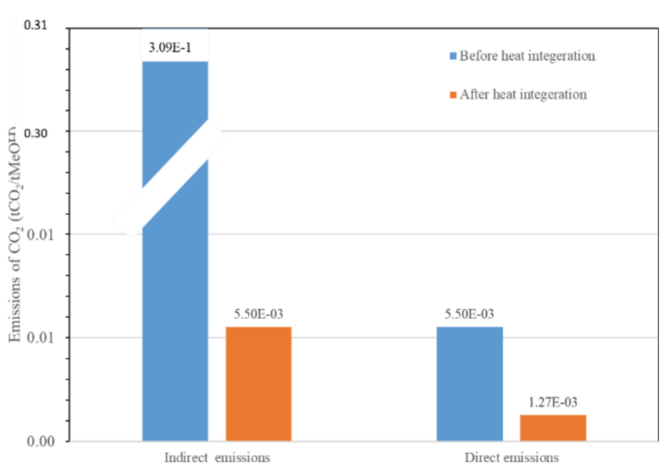

Fig. 8. Total estimated CO_2 emissions in t_{CO_2}/t_{MeOH} for CO_2 hydrogenation to methanol process before and after heat integration.

emissions were reduced by around 98% after conducting heat integration since the process streams provided sufficient duty for the heaters.

Additionally, when considering the optimized process, the estimated direct and indirect emissions are 98.6% and 96%, respectively, less than the values reported by [37], who reported direct emissions of 0.090 tCO_2/t_{MeOH} and indirect emissions of 0.136 tCO_2/t_{MeOH} for a European bio-methanol plant. The reduction in emissions is mainly attributed to the requirement of a smaller number of compressors and heat exchangers in this process for preparing CO_2 feed to feed specifications. This reflects the added value of incorporating a CO_2 to methanol process within the biomass supply chain for sustainable bio-methanol production.

In the presented process, high-pressure liquid and pure CO_2 were considered for methanol utilization. CO_2 captured from flue gases is a potential feed that will not impact the thermodynamic properties of the primary catalytic conversion process. However, additional CO_2 capture, treatment, and compression units will be needed to meet feed specifications. Hence, overall plant design and economic feasibility should be investigated. On the other hand, it is worth mentioning that different production routes can be utilized for CO_2 to methanol production, including CO_2 electrochemical reduction to methanol and two-step CO_2 catalytic conversion to methanol. In the latter route, CO_2 is first converted to CO, and CO is then hydrogenated to methanol in the second step. Both processes are still industrially immature and require further development. Moreover, despite the maturity of the CO_2 hydrogenation to methanol technology, catalyst development is still a dynamic area of research where other studies investigated the utilization of Ni/Ga [33,60], ZnO/ZrO_2 [61], and $InOx/ZrO_2$ [62] catalysts. The studied catalysts are still under development and have not reached the stage of industrial commercialization yet.

7. Conclusion and future considerations

A state of art of catalytic conversion of CO_2 to methanol is presented in this study. The economic and environmental feasibility of the proposed process under optimized operating conditions was explored. In comparison with previous studies, the assessed process involves less equipment due to the utilization of high-pressure and pure liquid CO_2, produced or generated from a former cryogenic biogas separation process or petrochemical industries. Optimized CO_2/H_2 feed ratio of 1:7 to achieve an overall CO_2 process conversion of 99% and methanol yield $\geq$ 99%. Simulation results indicated better performance for the adiabatic reactor than the isothermal reactor, with a reduced residence time of 48.46% and operating conditions of 210 °C. Overall energy efficiency was further improved by lowering external utilities by 63% after using the heat integration approach. Similar to the economic evaluation, which concluded that the process profitability is highly dependent on H_2

supply price, in this analysis, the financial assessment demonstrated the requirement of a maximum H_2 supply price of \$0.97/kg to break even the NPV for a 20-year project lifetime in the Middle East with annual methanol production of 2.34 kt/yr. From an environmental perspective, the optimized process successfully contributes to reducing total CO_2 emissions by 97.8% compared to the baseline process configuration. The catalyst $Cu/ZnO/Al_2O_3$ showed excellent efficiency for the industrial commercialization of CO_2 hydrogenation to methanol. Future research on process configuration and simulation could involve testing the efficiency and stability of different novel Cu, Pd, or Zn-based catalysts for CO_2 hydrogenation to methanol under varied operating conditions and CO_2/H_2 feed ratios.

CRediT authorship contribution statement

Noor Yusuf: Methodology, Software, Data curation, Writing – review & editing, Writing – original draft. **Fares Almomani:** Resources, Supervision, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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